HARUN ERKAM BALI

Backend Engineer | Python | Django | FastAPI | Go

Türkiye | Open to Remote • +90 539 569 28 75 • admin@harunerkam.com • harunerkam.com • linkedin.com/in/harunerkam • github.com/erkamspyjamas

PROFESSIONAL SUMMARY

Backend-focused software engineer with 3+ years of experience designing, building and operating production systems. Strong in Python, Django, Django REST Framework, FastAPI and Go, with hands-on experience in PostgreSQL, Redis, async/background processing, service integrations and real-time communication. Experienced in relational data modeling, API architecture, authentication/authorization, query optimization and production reliability. Also owns the operational side of backend delivery through Docker, Kubernetes/k3s, Nginx, PgBouncer, CI/CD, health checks, logging and monitoring.

CORE TECHNICAL SKILLS

Backend & Languages: Python, Go, TypeScript/JavaScript, Django, Django REST Framework, FastAPI, Node.js, Express.js, NestJS

API & Architecture: REST APIs, JWT/session auth, RBAC, permissions, validation, pagination, throttling/rate limiting, background jobs, queues, retries, timeouts, service integration, streaming gateways

Data & Caching: PostgreSQL, Redis, MySQL, MSSQL, MongoDB, relational modeling, schema design, migrations, indexing, query/ORM optimization, transactions, caching, connection pooling

Production & DevOps: Docker, Docker Compose, Kubernetes/k3s, Nginx, Traefik, PgBouncer, Gunicorn/Uvicorn, Linux, GitHub Actions, CI/CD, SSL/TLS, health/readiness checks, structured logging, monitoring

R&D / AI Systems: PyTorch, NVIDIA NeMo, ESPnet, VITS, TTS pipelines, inference/service integration, low-latency backend communication

PROFESSIONAL EXPERIENCE

Full Stack Developer & DevOps Engineer - R&D
Aug 2026 - Present

Neurosound Dikkat Ve Algı Geliştirme Merkezi | Türkiye

- Design backend and distributed-system architectures using Django, FastAPI, Go, PostgreSQL and Redis for R&D-focused production systems.
- Integrate AI/TTS, streaming and external-service pipelines with application backends, focusing on latency, reliability, failure handling and scalability.
- Prototype and evaluate service boundaries, API flows, data paths and deployment approaches before moving solutions into production.

Full Stack Developer & DevOps Engineer
Aug 2024 - Aug 2026

Neurosound Dikkat Ve Algı Geliştirme Merkezi | Türkiye

- Built and maintained backend services, REST APIs and internal systems using Django, FastAPI, Express.js/Node.js and Go.
- Designed PostgreSQL schemas and migrations; optimized ORM/SQL queries, indexes, caching strategies and PgBouncer-backed connection handling.
- Implemented authentication, permissions, service integrations, background-processing patterns and Redis-backed workloads for production applications.
- Deployed and operated services with Docker and k3s/Kubernetes behind Nginx; maintained CI/CD, SSL/TLS, ingress, health/readiness checks, logging and monitoring.

Full Stack Developer
Feb 2024 - Aug 2024

Neurosound Dikkat Ve Algı Geliştirme Merkezi | Türkiye

- Developed web backends and REST APIs using Django, FastAPI and Express.js, including authentication systems, internal services and third-party integrations.
- Designed relational data models and reusable backend modules, and improved PostgreSQL/Redis data-access and caching patterns.
- Built automation tools and operational workflows that reduced repetitive manual work and improved consistency.

Software Developer
Nov 2023 - Feb 2024

Neurosound Dikkat Ve Algı Geliştirme Merkezi | Türkiye

- Developed an interactive educational application in Unity/C#, including application logic, data handling, reusable components, testing and debugging.
- Worked closely with product/education requirements and translated them into maintainable software flows and user-facing functionality.

EARLIER EXPERIENCE

Research & Development Engineer **Feb 2023 - Jun 2023**

AutoDidactic Technologies | Konya, Türkiye

- Built Python and Node.js prototypes for AI/R&D experiments, data-processing workflows and technical reporting.
- Supported Docker-based development, Linux administration, AWS EC2 deployments and automation scripts.
- Contributed to reinforcement-learning and experimental software work with an emphasis on rapid iteration and reproducibility.

SELECTED BACKEND PROJECTS

Sustindex - Backend Architecture: Designed backend services for a FinTech platform using Django, FastAPI and PostgreSQL, including API design, authentication, relational data modeling and production-oriented deployment patterns.

Realtime TTS Serving / Integration: Integrated application backends with Python-based TTS workers and Go/Python service layers, focusing on low-latency communication, timeouts, retries, worker orchestration and reliable service-to-service flows.

Django Production Platform: Built multi-app Django systems with role-aware access, dashboards, media/static handling, PostgreSQL optimization, Redis caching and Nginx/Gunicorn/Docker deployment.

Stock & Device Tracking Backend: Implemented inventory, cargo and device-management services using Node.js/Express.js with Firebase and MSSQL variants, including data validation and operational workflows.

AI / TTS Training Workflow Services: Worked with NVIDIA NeMo, ESPnet and PyTorch-based tooling around dataset preparation, training/fine-tuning workflows, artifacts and backend service integration.

EDUCATION

B.Sc., Management Information Systems - Necmettin Erbakan University

2019 - 2023 | GPA: 3.11 / 4.00

CERTIFICATIONS

- EF SET English Certificate - 65/100 (C1 Advanced)
- Aygaz Machine Learning Bootcamp

LANGUAGES

Turkish: Native | English: C1 (EF SET 65/100)

Selected repositories and production systems are private; architecture overviews, code excerpts or access can be shared when appropriate.